

MANITOWOC COUNTY AP, WI, USA

WMO: 726455

Lat: 44.133N

Lon: 87.667W

Elev: 651

StdP: 14.35

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94897

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.9	0.0	-16.3	2.3	-2.3	-10.1	3.3	1.0	26.5	18.3	24.4	20.6	11.4	280	0.587

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	85.0	71.8	82.0	70.5	79.8	68.8	74.9	82.7	72.8	79.8	70.9	77.3	11.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.3	122.7	80.2	70.4	114.5	77.5	68.7	107.9	75.5	38.8	83.0	36.7	79.6	35.1	77.5	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.9	20.6	18.7		-10.0	91.1	6.1	4.1	-14.4	94.0	-18.0	96.5	-21.5	98.8	-25.9	101.8	
				-10.9	78.2	5.2	2.1	-14.6	79.7	-17.6	80.9	-20.5	82.0	-24.3	83.5	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.4	20.3	22.6	31.9	42.6	53.3	63.6	69.0	67.6	60.8	49.1	37.1	25.8	
	DBStd	19.03	11.46	10.85	9.84	7.53	7.28	6.70	5.72	5.43	7.02	7.83	8.65	10.55	
	HDD50	3806	922	767	565	243	47	1	0	0	4	113	392	752	
	HDD65	7512	1387	1187	1026	672	370	105	25	33	161	494	836	1216	
	CDD50	2141	0	0	4	21	149	407	590	547	330	87	6	0	
	CDD65	372	0	0	0	0	6	62	149	115	37	3	0	0	
	CDH74	2446	0	0	1	9	80	473	1066	617	184	16	0	0	
	CDH80	557	0	0	0	1	13	108	285	117	31	2	0	0	
Wind	WSAvg	8.8	10.4	10.2	9.5	9.7	8.6	7.3	7.1	6.9	7.7	9.0	9.7	9.7	
Precipitation	PrecAvg	29.2	1.6	1.2	1.8	2.8	3.0	3.2	3.2	3.3	3.1	2.2	2.3	1.7	
	PrecMax	44.6	10.2	2.8	5.2	7.6	8.7	8.5	8.1	9.5	12.6	5.6	6.2	4.1	
	PrecMin	20.8	0.1	0.0	0.0	0.6	0.6	0.4	0.8	0.2	0.2	0.2	0.1	0.1	
	PrecStd	5.0	1.6	0.7	1.3	1.4	1.7	1.8	1.5	1.9	2.2	1.1	1.4	1.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.4	48.5	63.6	73.2	81.5	88.0	91.3	88.2	84.2	75.4	63.9	52.4	
		MCWB	40.8	42.5	56.0	59.7	66.4	73.5	76.1	74.6	71.9	64.6	54.7	49.1	
	2%	DB	41.0	43.2	54.5	63.8	75.1	82.5	86.1	82.5	79.2	69.7	56.9	46.1	
		MCWB	37.5	38.3	47.7	53.2	63.1	69.8	73.1	71.3	68.8	61.4	50.0	42.4	
	5%	DB	37.3	39.1	49.5	58.1	71.0	79.4	82.3	80.6	75.1	64.4	52.6	42.9	
		MCWB	35.2	35.9	43.9	48.6	60.2	67.6	71.4	69.9	66.6	58.1	47.5	39.6	
	10%	DB	36.0	36.7	45.0	54.3	66.1	75.3	80.6	77.3	72.5	61.5	49.1	39.1	
		MCWB	34.0	34.0	40.2	46.2	57.3	65.5	70.1	68.1	65.3	56.0	45.3	36.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.3	44.2	57.1	60.8	68.5	75.6	78.2	77.2	73.5	67.8	56.6	50.3
			MCDB	45.5	47.6	63.0	72.1	78.1	84.9	88.1	85.0	80.6	73.6	62.3	51.8
2%		WB	38.1	39.3	49.0	54.9	65.1	71.9	75.6	73.9	70.8	63.0	51.6	43.8	
		MCDB	40.1	42.7	53.5	62.9	73.3	80.4	83.6	80.8	76.7	66.6	55.3	45.6	
5%		WB	35.9	36.6	44.2	50.3	61.6	69.3	73.4	71.7	68.7	59.5	48.3	40.0	
		MCDB	37.7	39.0	48.4	56.6	69.2	77.4	80.7	77.4	73.6	64.0	51.8	42.4	
10%		WB	33.4	34.3	40.6	46.8	58.2	66.9	71.4	69.9	66.7	56.5	45.7	36.9	
		MCDB	35.6	36.7	44.8	52.9	64.9	74.2	78.0	75.3	71.6	60.9	48.7	39.2	
Mean Daily Temperature Range	5% DB	MDBR	12.7	13.7	13.8	15.3	17.2	17.4	17.1	15.7	16.6	14.6	12.8	12.0	
		MCDBR	13.5	13.8	19.6	23.5	24.7	21.8	19.8	17.9	19.5	19.4	17.9	14.2	
		MCWBR	11.9	11.3	14.5	15.3	15.3	12.7	11.0	10.2	12.1	13.4	13.7	12.1	
	5% WB	MCDBR	12.7	13.0	19.0	22.3	22.7	20.4	18.2	16.1	17.2	17.5	16.2	13.1	
		MCWBR	11.7	11.1	15.0	15.8	15.7	13.4	11.5	10.3	11.6	13.5	13.5	11.9	
Clear-Sky Solar Irradiance	taub	0.278	0.310	0.332	0.380	0.403	0.418	0.408	0.403	0.379	0.338	0.305	0.280		
	taud	2.440	2.338	2.391	2.301	2.289	2.293	2.326	2.336	2.388	2.497	2.519	2.486		
	Ebn at Noon	272	279	287	280	276	271	272	270	268	267	259	260		
	Edh at Noon	23	30	33	39	41	41	39	37	33	26	21	20		
All-Sky Solar Radiation	RadAvg	469	777	1149	1429	1715	1888	1936	1654	1311	826	513	365		
	RadStd	59	81	107	137	141	134	84	108	104	86	47	38		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page